



Bioinformatics

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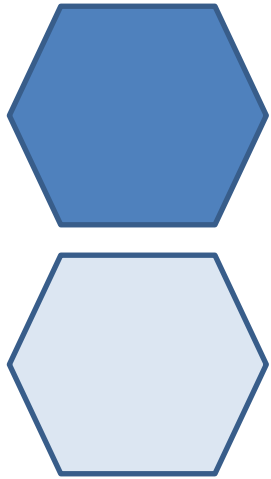
 Hands on KEGG and Biomart

Exercise KEGG

- ▶ Given the gene list named `entrezID.txt`
 1. Go on KEGG pathways database
 2. Choose homo sapiens cell cycle as pathway
 3. Highlight genes of the list that are on cell cycle pathway as yellow boxes

Exercise Biomart

- ▶ Given the gene list named `geneSymbol.txt`
 1. Upload this list in Biomart
 2. Convert the identifiers from HGNC Official Gene symbols to Entrez Gene ID
 3. Indicate which are the associated GO slim terms
 4. Indicate if they are protein coding or non-coding (gene type)



Solutions KEGG



KEGG PATHWAY Database

Wiring diagrams of molecular interactions, reactions and relations

Menu PATHWAY BRITE MODULE KO GENES LIGAND NETWORK DISEASE DRUG DBGET

Select prefix

hsa

Organism

Enter keywords

cell cycle

Go

Help

[[New pathway maps](#) | [Update history](#)]

Pathway Maps

KEGG PATHWAY is a collection of manually drawn [pathway maps](#) representing our knowledge on the molecular interaction, reaction and relation networks for:

1. Metabolism

[Global/overview](#) [Carbohydrate](#) [Energy](#) [Lipid](#) [Nucleotide](#) [Amino acid](#) [Other amino](#) [Glycan](#)
[Cofactor/vitamin](#) [Terpenoid/PK](#) [Other secondary metabolite](#) [Xenobiotics](#) [Chemical structure](#)

2. Genetic Information Processing

3. Environmental Information Processing

4. Cellular Processes

5. Organismal Systems

6. Human Diseases

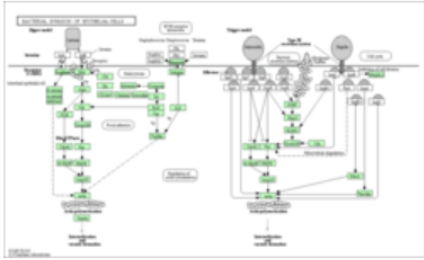
7. Drug Development

KEGG PATHWAY is the reference database for pathway mapping in [KEGG Mapper](#).

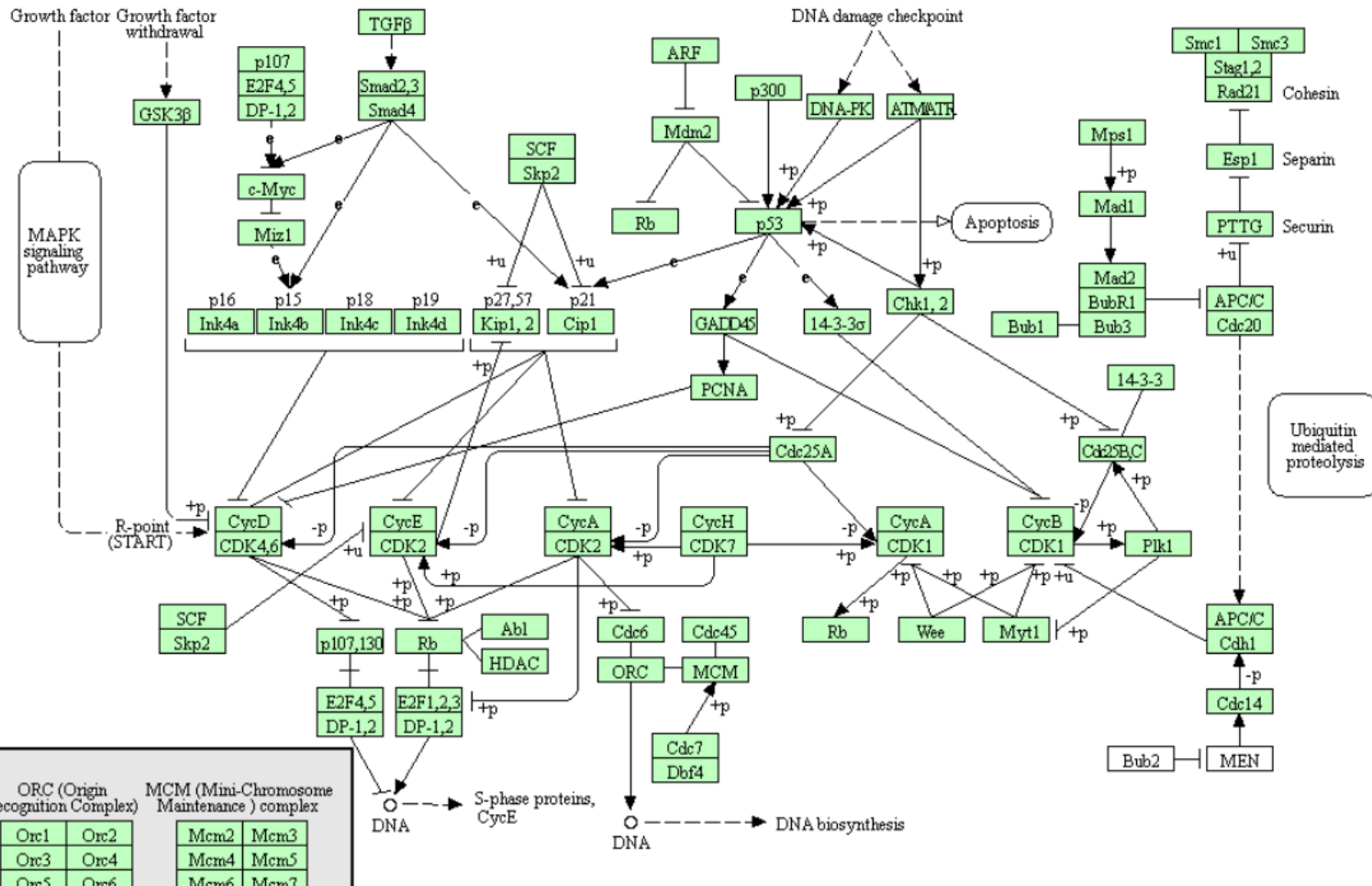
Pathway Text Search

Number of entries in a page

Page : of 5 Items : 1 - 20 of 81

Entry	Thumbnail Image	Name	Description	Object
hsa04110		Cell cycle - Homo sapiens (human)	Mitotic cell cycle progression is accomplished through a reproducible sequence of events, DNA replic...	...ysis hsa04010: MAPK signaling pathway hsa04110: Cell cycle
hsa05100		Bacterial invasion of epithelial cells - Homo sapiens (human)	...cteria can invade phagocytic and non-phagocytic cells and colonize them intracell ^u larly, then become...	...MO1) hsa05100: Bacterial invasion of epithelial cells hsa04512: ECM-receptor interaction hsa04810: R...

CELL CYCLE



Entrez ID [tab] background color

Prepare your list



Copy and paste it on KEGG
Color mapping panel

	A	B
1	4363	yellow
2	26027	yellow
3	23305	yellow
4	113146	yellow
5	124930	yellow
6	338440	yellow
7	306	yellow
8	54848	yellow
9	80210	yellow
10	9915	yellow
11	55723	yellow
12	440	yellow

▼ Option

Scale: 100%

▼ Search

Go

▼ ID search

Go

▼ Color

▼ Network

- ☐ nt06130 Cell cycle (viruses)
- ☐ N00536 MDM2-p21-Cell cycle
- ☐ N00499 ATR-p21-Cell cycle G2
- ☐ N00493 Spindle assembly che
- ☐ nt06230 Cell cycle
- ☐ N00066 MDM2-p21-Cell cycle
- ☐ N00090 p15-Cell cycle G1/S
- ☐ N00091 p27-Cell cycle G1/S

Color

Enter objects one per line followed by color specification:

4363 yellow
26027 yellow
23305 yellow
113146 yellow
124930 yellow
338440 yellow
306 yellow

Scegli file Nessun file selezionato

☐ Numerical values converted to color gradation

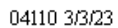
Enter dataset name (option):

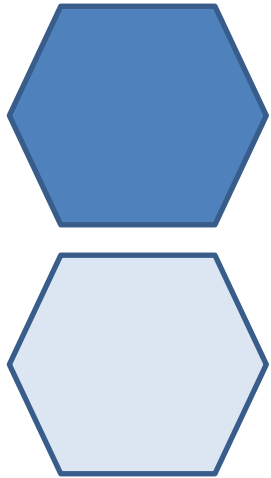
☐ Apply to all maps

Exec Close Apply

If it doesn't work insert a space
and not a tab between the node
name and the bg color

CELL CYCLE





Solutions Biomart

GO on Biomart

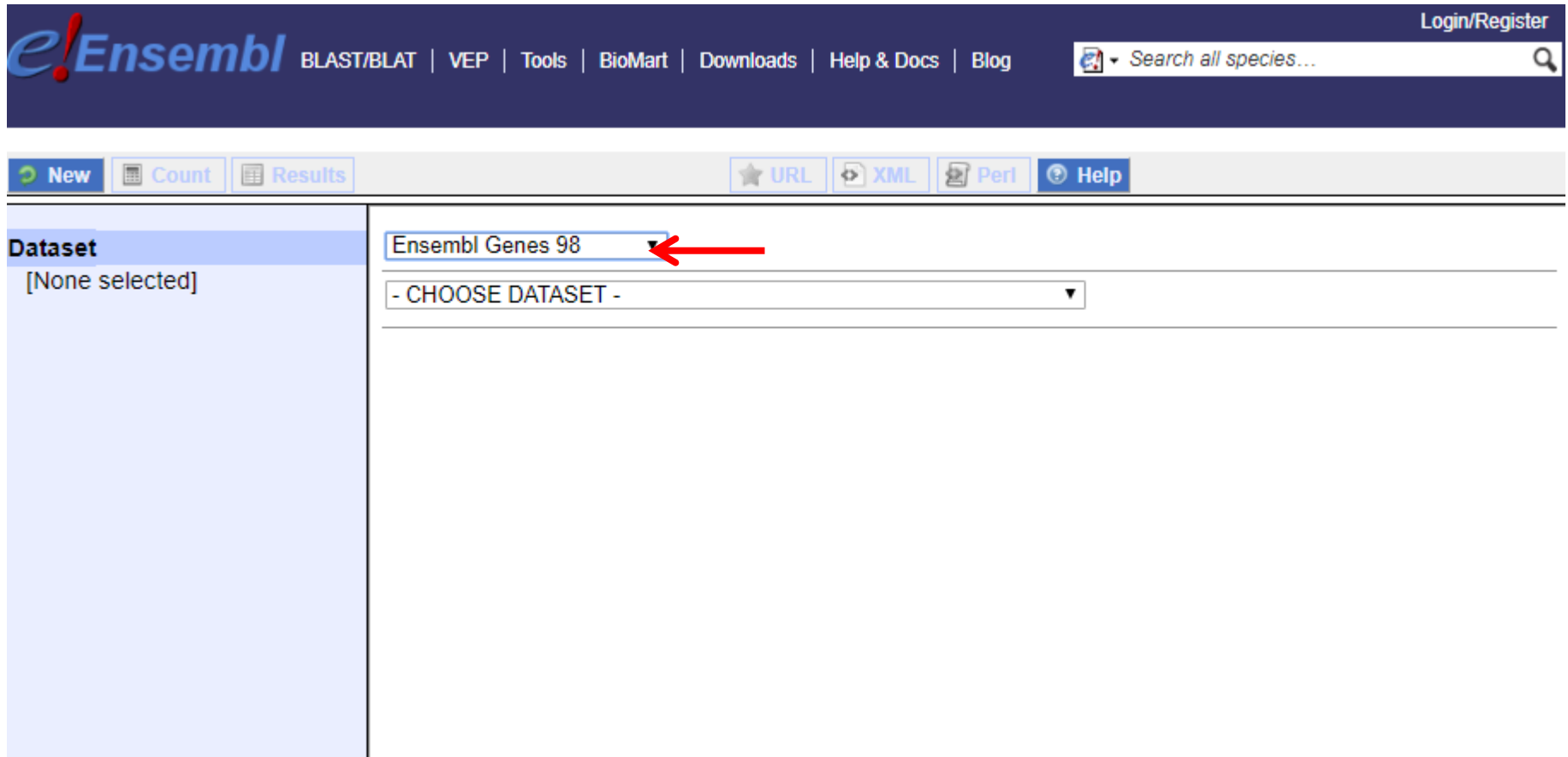
<http://www.ensembl.org/biomart/martview/e7dc3889bae0ff3d34cb45dacf0a1b0c>

The screenshot displays the Ensembl BioMart web interface. At the top, the Ensembl logo is on the left, and navigation links for BLAST/BLAT, BioMart, Tools, Downloads, Help & Documentation, Blog, and Mirrors are in the center. On the right, there is a 'Login/Register' link and a search bar with the placeholder text 'Search all species...'. Below the navigation bar, a secondary bar contains buttons for 'New', 'Count', 'Results', 'URL', 'XML', 'Perl', and 'Help'. The main content area is divided into two sections. On the left, under the heading 'Dataset', it says '[None selected]'. On the right, there is a large white box with a dropdown menu at the top that reads '- CHOOSE DATABASE -'. The rest of the box is empty.



Choose Database

► Ensembl genes 98



The screenshot shows the Ensembl genome browser interface. At the top is a dark blue header with the Ensembl logo, navigation links (BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, Blog), a search bar labeled "Search all species...", and a "Login/Register" link. Below the header is a light gray toolbar with buttons for "New", "Count", "Results", "URL", "XML", "Perl", and "Help". The main content area is divided into two panels. The left panel, titled "Dataset", shows "[None selected]". The right panel contains a dropdown menu with "Ensembl Genes 98" selected, indicated by a red arrow. Below this is another dropdown menu labeled "- CHOOSE DATASET -".

Choose dataset

► Homo sapiens hg38

The screenshot displays the Ensembl genome browser interface. At the top, the Ensembl logo is on the left, and navigation links (BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, Blog) and a search bar (Search all species...) are on the right. A 'Login/Register' link is also present. Below the header, a secondary navigation bar contains buttons for 'New', 'Count', 'Results', 'URL', 'XML', 'Perl', and 'Help'. The main content area is divided into a left sidebar and a right pane. The sidebar has sections for 'Dataset' (showing 'Human genes (GRCh38.p13)'), 'Filters' (showing '[None selected]'), 'Attributes' (listing 'Gene stable ID', 'Gene stable ID version', 'Transcript stable ID', and 'Transcript stable ID version'), and another 'Dataset' section (showing '[None Selected]'). The right pane features two dropdown menus. The top dropdown is labeled 'Ensembl Genes 98'. The bottom dropdown is labeled 'Human genes (GRCh38.p13)' and is highlighted with a red arrow, indicating the selected dataset.

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog Login/Register

Search all species...

New Count Results URL XML Perl Help

Dataset
Human genes (GRCh38.p13)

Filters
[None selected]

Attributes
Gene stable ID
Gene stable ID version
Transcript stable ID
Transcript stable ID version

Dataset
[None Selected]

Ensembl Genes 98

Human genes (GRCh38.p13)

Click **Filters** → GENE to load input list

e!Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog Search all species... Login/Register

New Count Results URL XML Perl Help

Dataset
Human genes (GRCh38.p13)
Filters ←
[None selected]
Attributes
Gene stable ID
Gene stable ID version
Transcript stable ID
Transcript stable ID version

Dataset
[None Selected]

Please restrict your query using criteria below
(If filter values are truncated in any lists, hover over the list item to see the full text)

☒ **REGION:**

☒ **GENE:** ←

☐ Limit to genes (external references)...
With CCDS ID(s)
● Only
○ Excluded

☐ Input external references ID list [Max 500 advised]
Gene stable ID(s) [e.g. ENSG000000000003]
Scegli file Nessun file selezionato

☐ Limit to genes (microarray probes/probesets)...
With AFFY HC G110 probe ID(s)
● Only
○ Excluded

☐ Input microarray probes/probesets ID list [Max 500 advised]
AFFY HC G110 probe ID(s) [e.g. 1787_at]

Check Input external reference ID list

and load the gene list to be converted (HGNC symbol in gene_list.txt)

e!Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog Login/Register

Search all species...

New Count Results URL XML Perl Help

Dataset
Human genes (GRCh38.p13)

Filters
Gene stable ID(s) [e.g. ENSG000000000003]: [ID-list specified]

Attributes
Gene stable ID
Gene stable ID version
Transcript stable ID
Transcript stable ID version

Dataset
[None Selected]

Please restrict your query using criteria below
(If filter values are truncated in any lists, hover over the list item to see the full text)

☐ REGION:

☐ GENE:

☐ Limit to genes (external references)...
With CCDS ID(s)
☒ Only
☐ Excluded

☒ Input external references ID list [Max 500 advised]

Gene stable ID(s) [e.g. ENSG000000000003]

Scegli file lista_geni.txt

☐ Limit to genes (microarray probes/probesets)...
With AFFY HC G110 probe ID(s)
☒ Only
☐ Excluded

☐ Input microarray probes/probesets ID list [Max 500 advised]

AFFY HC G110 probe ID(s) [e.g. 1787_at]

Choose **HGNC symbols(s)** as ID type of the input gene list

e!Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog Search all species... Login/Register

New **Count** **Results** **URL** **XML** **Perl** **Help**

Dataset
Human genes (GRCh38.p13)

Filters
HGNC symbol(s) [e.g. A1BG]:
[ID-list specified]

Attributes
Gene stable ID
Gene stable ID version
Transcript stable ID
Transcript stable ID version

Dataset
[None Selected]

Please restrict your query using criteria below
(If filter values are truncated in any lists, hover over the list item to see the full text)

REGION:

GENE:

☐ Limit to genes (external references)...
With CCDS ID(s) ☐ Only ☐ Excluded

☒ Input external references ID list [Max 500 advised]
HGNC symbol(s) [e.g. A1BG]
Scegli file lista_geni.txt

☐ Limit to genes (microarray probes/probesets)...
With AFFY HC G110 probe ID(s) ☐ Only ☐ Excluded

☐ Input microarray probes/probesets ID list [Max 500 advised]
AFFY HC G110 probe ID(s) [e.g. 1787_at]
Scegli file Nessun file selezionato

Click [Attributes](#) to select your output

The screenshot shows the Ensembl BioMart interface. At the top, the Ensembl logo is on the left, and navigation links (BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, Blog) and a search bar are on the right. Below the navigation bar, there are tabs for 'New', 'Count', and 'Results'. A secondary bar contains links for 'URL', 'XML', 'Perl', and 'Help'. The left sidebar is divided into sections: 'Dataset' (Human genes (GRCh38.p13)), 'Filters' (HGNC symbol(s) [e.g. A1BG]: [ID-list specified]), and 'Attributes' (highlighted with a red arrow). Under 'Attributes', options include 'Gene stable ID', 'Gene stable ID version', 'Transcript stable ID', and 'Transcript stable ID version'. The main content area has a heading 'Please select columns to be included in the output and hit 'Results' when ready' and a note about missing non-coding genes. It contains a list of attributes with radio buttons: 'Features' (selected), 'Structures', 'Homologues', 'Variant (Germline)', 'Variant (Somatic)', and 'Sequences'. Below this are three expandable sections: 'GENE:', 'EXTERNAL:', and 'PROTEIN DOMAINS AND FAMILIES:'.

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog Login/Register

Search all species...

New Count Results URL XML Perl Help

Dataset
Human genes (GRCh38.p13)

Filters
HGNC symbol(s) [e.g. A1BG]:
[ID-list specified]

Attributes
Gene stable ID
Gene stable ID version
Transcript stable ID
Transcript stable ID version

Dataset
[None Selected]

Please select columns to be included in the output and hit 'Results' when ready

Missing non coding genes in your mart query output, please check the following [FAQ](#)

☒ **Features** ☐ **Variant (Germline)**
☐ **Structures** ☐ **Variant (Somatic)**
☐ **Homologues** ☐ **Sequences**

☐ GENE:

☐ EXTERNAL:

☐ PROTEIN DOMAINS AND FAMILIES:

Click **Attributes** to select your output

I. Gene type (GENE menu)

The screenshot shows the Ensembl genome browser interface. The top navigation bar includes links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog. A search bar is present on the right. The left sidebar shows the 'Attributes' menu with 'Gene type' selected. The main area displays a list of attributes under the 'GENE' section, with 'Gene type' checked. A red arrow points to the 'Gene type' checkbox.

Please select columns to be included in the output and hit 'Results' when ready

Missing non coding genes in your mart query output, please check the following [FAQ](#)

☒ Features ☐ Variant (Germline)
☐ Structures ☐ Variant (Somatic)
☐ Homologues ☐ Sequences

☒ GENE:

Ensembl

- ☐ Gene stable ID
- ☐ Gene stable ID version
- ☐ Transcript stable ID
- ☐ Transcript stable ID version
- ☐ Protein stable ID
- ☐ Protein stable ID version
- ☐ Exon stable ID
- ☐ Gene description
- ☐ Chromosome/scaffold name
- ☐ Gene start (bp)
- ☐ Gene end (bp)
- ☐ Strand
- ☐ Karyotype band
- ☐ Transcript start (bp)
- ☐ Transcript support level (TSL)
- ☐ GENCODE basic annotation
- ☐ APPRIS annotation
- ☐ RefSeq match transcript
- ☐ Gene name
- ☐ Source of gene name
- ☐ Transcript name
- ☐ Source of transcript name
- ☐ Transcript count
- ☐ Gene % GC content
- ☒ Gene type
- ☐ Transcript type
- ☐ Source (gene)
- ☐ Source (transcript)

Click **Attributes** to select your output

2. **Go slim** GOA Accessions and description (EXTERNAL menu)

Dataset
Human genes (GRCh38.p13)

Filters
HGNC symbol(s) [e.g. A1BG]:
[ID-list specified]

Attributes
Gene type
GOSlim GOA Accession(s)
GOSlim GOA Description

Dataset
[None Selected]

Please select columns to be included in the output and hit 'Results' when ready

Missing non coding genes in your mart query output, please check the following [FAQ](#)

☐ Features ☐ Variant (Germline)
☐ Structures ☐ Variant (Somatic)
☐ Homologues ☐ Sequences

☐ GENE:

☐ EXTERNAL:

GO

☐ GO term accession ☐ GO term evidence code
☐ GO term name ☐ GO domain
☐ GO term definition

GOSlim GOA

☒ GOSlim GOA Accession(s) ☒ GOSlim GOA Description

External References (max 3)

☐ CCDS ID ☐ miRBase ID
☐ ChEMBL ID ☐ miRBase transcript name ID
☐ Clone-based (Ensembl) gene ID ☐ NCBI gene description
☐ Clone-based (Ensembl) transcript ID ☐ NCBI gene accession
☐ DataBase of Aberrant 3' Splice Sites name ☐ NCBI gene ID

Click [Attributes](#) to select your output

3. [NCBI](#) gene id (EXTERNAL menu) which corresponds to the Entrez id

The screenshot shows the Ensembl genome browser interface. The top navigation bar includes the Ensembl logo, links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog, along with a search bar and a Login/Register link. The main content area is divided into several sections. On the left, the 'Dataset' section shows 'Human genes (GRCh38.p13)' and the 'Attributes' section lists various attributes including 'Gene type', 'GOSlim GOA Accession(s)', 'GOSlim GOA Description', and 'NCBI gene ID'. The 'External References' tab is selected, displaying a list of external database identifiers. The 'NCBI gene ID' checkbox is checked, and a red arrow points to it. Another red arrow points to the 'Attributes' tab in the left sidebar.

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog Login/Register

Search all species...

New Count Results URL XML Perl Help

Dataset
Human genes (GRCh38.p13)

Filters
HGNC symbol(s) [e.g. A1BG]:
[ID-list specified]

Attributes
Gene type
GOSlim GOA Accession(s)
GOSlim GOA Description
NCBI gene ID

External References (max 3)

- ☐ CCDS ID
- ☐ ChEMBL ID
- ☐ Clone-based (Ensembl) gene ID
- ☐ Clone-based (Ensembl) transcript ID
- ☐ DataBase of Aberrant 3' Splice Sites name
- ☐ DataBase of Aberrant 3' Splice Sites ID
- ☐ DataBase of Aberrant 5' Splice Sites name
- ☐ DataBase of Aberrant 5' Splice Sites ID
- ☐ Ensembl Human Transcript ID
- ☐ Ensembl Human Translation ID
- ☐ EntrezGene transcript name ID
- ☐ European Nucleotide Archive ID
- ☐ Expression Atlas ID
- ☐ GeneDB ID
- ☐ HGNC ID
- ☐ HGNC symbol
- ☐ Human Protein Atlas accession
- ☐ Human Protein Atlas ID
- ☐ INSDC protein ID
- ☐ KEGG Pathway and Enzyme ID
- ☐ LRG display in Ensembl gene ID
- ☐ LRG display in Ensembl transcript ID
- ☐ MEROPS - the Peptidase Database ID
- ☐ miRBase ID
- ☐ miRBase transcript name ID
- ☐ NCBI gene description
- ☐ NCBI gene accession
- ☒ NCBI gene ID
- ☐ PDB ID
- ☐ Reactome ID
- ☐ Reactome gene ID
- ☐ Reactome transcript ID
- ☐ RefSeq mRNA ID
- ☐ RefSeq mRNA predicted ID
- ☐ RefSeq ncRNA ID
- ☐ RefSeq ncRNA predicted ID
- ☐ RefSeq peptide ID
- ☐ RefSeq peptide predicted ID
- ☐ RFAM ID
- ☐ RFAM transcript name ID
- ☐ RNACentral ID
- ☐ Transcript name ID
- ☐ UCSC Stable ID
- ☐ UniParc ID
- ☐ UniProtKB Gene Name symbol
- ☐ UniProtKB Gene Name ID

Click [Attributes](#) to select your output

4. [HGNC symbols](#) (EXTERNAL menu) which corresponds to what you gave as input.

The screenshot shows the Ensembl genome browser interface. At the top, the Ensembl logo is on the left, and navigation links (BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, Blog) and a search bar are on the right. Below the header, there are tabs for 'New', 'Count', and 'Results'. A secondary row of tabs includes 'URL', 'XML', 'Perl', and 'Help'. The main content area is divided into a left sidebar and a central panel. The sidebar has sections for 'Dataset' (Human genes (GRCh38.p13)), 'Filters' (HGNC symbol(s) [e.g. A1BG]: [ID-list specified]), and 'Attributes'. The 'Attributes' section lists various identifiers: Gene type, GOSlim GOA Accession(s), GOSlim GOA Description, NCBI gene ID, and HGNC symbol. The 'Dataset' section at the bottom of the sidebar shows '[None Selected]'. A red arrow points from this section to the 'HGNC symbol' option in the central panel. The central panel, titled 'External References (max 3)', contains a list of checkboxes for various identifiers. The 'HGNC symbol' checkbox is checked. Other visible options include CCDS ID, ChEMBL ID, Clone-based (Ensembl) gene ID, Clone-based (Ensembl) transcript ID, DataBase of Aberrant 3' Splice Sites name, DataBase of Aberrant 3' Splice Sites ID, DataBase of Aberrant 5' Splice Sites name, DataBase of Aberrant 5' Splice Sites ID, Ensembl Human Transcript ID, Ensembl Human Translation ID, EntrezGene transcript name ID, European Nucleotide Archive ID, Expression Atlas ID, GeneDB ID, HGNC ID, Human Protein Atlas accession, Human Protein Atlas ID, INSDC protein ID, KEGG Pathway and Enzyme ID, LRG display in Ensembl gene ID, LRG display in Ensembl transcript ID, MEROPS - the Peptidase Database ID, miRBase ID, miRBase transcript name ID, NCBI gene description, NCBI gene accession, NCBI gene ID, PDB ID, Reactome ID, Reactome gene ID, Reactome transcript ID, RefSeq mRNA ID, RefSeq mRNA predicted ID, RefSeq ncRNA ID, RefSeq ncRNA predicted ID, RefSeq peptide ID, RefSeq peptide predicted ID, RFAM ID, RFAM transcript name ID, RNACentral ID, Transcript name ID, UCSC Stable ID, UniParc ID, UniProtKB Gene Name symbol, and UniProtKB Gene Name ID.

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog Login/Register

Search all species...

New Count Results URL XML Perl Help

Dataset
Human genes (GRCh38.p13)

Filters
HGNC symbol(s) [e.g. A1BG]:
[ID-list specified]

Attributes
Gene type
GOSlim GOA Accession(s)
GOSlim GOA Description
NCBI gene ID
HGNC symbol

Dataset
[None Selected]

External References (max 3)

- ☐ CCDS ID
- ☐ ChEMBL ID
- ☐ Clone-based (Ensembl) gene ID
- ☐ Clone-based (Ensembl) transcript ID
- ☐ DataBase of Aberrant 3' Splice Sites name
- ☐ DataBase of Aberrant 3' Splice Sites ID
- ☐ DataBase of Aberrant 5' Splice Sites name
- ☐ DataBase of Aberrant 5' Splice Sites ID
- ☐ Ensembl Human Transcript ID
- ☐ Ensembl Human Translation ID
- ☐ EntrezGene transcript name ID
- ☐ European Nucleotide Archive ID
- ☐ Expression Atlas ID
- ☐ GeneDB ID
- ☐ HGNC ID
- ☒ HGNC symbol
- ☐ Human Protein Atlas accession
- ☐ Human Protein Atlas ID
- ☐ INSDC protein ID
- ☐ KEGG Pathway and Enzyme ID
- ☐ LRG display in Ensembl gene ID
- ☐ LRG display in Ensembl transcript ID
- ☐ MEROPS - the Peptidase Database ID
- ☐ miRBase ID
- ☐ miRBase transcript name ID
- ☐ NCBI gene description
- ☐ NCBI gene accession
- ☒ NCBI gene ID
- ☐ PDB ID
- ☐ Reactome ID
- ☐ Reactome gene ID
- ☐ Reactome transcript ID
- ☐ RefSeq mRNA ID
- ☐ RefSeq mRNA predicted ID
- ☐ RefSeq ncRNA ID
- ☐ RefSeq ncRNA predicted ID
- ☐ RefSeq peptide ID
- ☐ RefSeq peptide predicted ID
- ☐ RFAM ID
- ☐ RFAM transcript name ID
- ☐ RNACentral ID
- ☐ Transcript name ID
- ☐ UCSC Stable ID
- ☐ UniParc ID
- ☐ UniProtKB Gene Name symbol
- ☐ UniProtKB Gene Name ID

Click Results

e!Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog Login/Register

Search all species...

New Count **Results** URL XML Perl Help

Dataset
Human genes (GRCh38.p13)

Filters **input**
HGNC symbol(s) [e.g. A1BG]:
[ID-list specified]

Attributes **output**
Gene type
GOSlim GOA Accession(s)
GOSlim GOA Description
NCBI gene ID
HGNC symbol

Dataset
[None Selected]

External References (max 3)

- ☐ CCDS ID
- ☐ ChEMBL ID
- ☐ Clone-based (Ensembl) gene ID
- ☐ Clone-based (Ensembl) transcript ID
- ☐ DataBase of Aberrant 3' Splice Sites name
- ☐ DataBase of Aberrant 3' Splice Sites ID
- ☐ DataBase of Aberrant 5' Splice Sites name
- ☐ DataBase of Aberrant 5' Splice Sites ID
- ☐ Ensembl Human Transcript ID
- ☐ Ensembl Human Translation ID
- ☐ EntrezGene transcript name ID
- ☐ European Nucleotide Archive ID
- ☐ Expression Atlas ID
- ☐ GeneDB ID
- ☐ HGNC ID
- ☒ HGNC symbol
- ☐ Human Protein Atlas accession
- ☐ Human Protein Atlas ID
- ☐ INSDC protein ID
- ☐ KEGG Pathway and Enzyme ID
- ☐ LRG display in Ensembl gene ID
- ☐ LRG display in Ensembl transcript ID
- ☐ MEROPS - the Peptidase Database ID
- ☐ miRBase ID
- ☐ miRBase transcript name ID
- ☐ NCBI gene description
- ☐ NCBI gene accession
- ☒ NCBI gene ID
- ☐ PDB ID
- ☐ Reactome ID
- ☐ Reactome gene ID
- ☐ Reactome transcript ID
- ☐ RefSeq mRNA ID
- ☐ RefSeq mRNA predicted ID
- ☐ RefSeq ncRNA ID
- ☐ RefSeq ncRNA predicted ID
- ☐ RefSeq peptide ID
- ☐ RefSeq peptide predicted ID
- ☐ RFAM ID
- ☐ RFAM transcript name ID
- ☐ RNAcentral ID
- ☐ Transcript name ID
- ☐ UCSC Stable ID
- ☐ UniParc ID
- ☐ UniProtKB Gene Name symbol
- ☐ UniProtKB Gene Name ID

View the first 10 rows

 BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Login/Register

 Search all species... 

New | Count | Results

★ URL | ▶ XML | 📄 Perl | ⓘ Help

Dataset
Human genes (GRCh38.p13)
Filters
HGNC symbol(s) [e.g. A1BG]:
[ID-list specified]
Attributes
Gene type
GOSlim GOA Accession(s)
GOSlim GOA Description
NCBI gene ID
HGNC symbol

Dataset
[None Selected]

Export all results to ☐ Unique results only

Email notification to

View rows as ☐ Unique results only

Gene type	GOSlim GOA Accession(s)	GOSlim GOA Description	NCBI gene ID	HGNC symbol
protein_coding	GO:0008150	biological process	4363	ABCC1
protein_coding	GO:0005575	cellular component	4363	ABCC1
protein_coding	GO:0022857	transmembrane transporter activity	4363	ABCC1
protein_coding	GO:0003674	molecular function	4363	ABCC1
protein_coding	GO:0055085	transmembrane transport	4363	ABCC1
protein_coding	GO:0006810	transport	4363	ABCC1
protein_coding	GO:0043167	ion binding	4363	ABCC1
protein_coding	GO:0005623	cell	4363	ABCC1
protein_coding	GO:0005886	plasma membrane	4363	ABCC1
protein_coding	GO:0016887	ATPase activity	4363	ABCC1

View ALL

 BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Login/Register

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New | Count | Results | URL | XML | Perl | Help

Dataset

Human genes (GRCh38.p13)

Filters

HGNC symbol(s) [e.g. A1BG]
[ID-list specified]

Attributes

Gene type
GOSlim GOA Accession(s)
GOSlim GOA Description
NCBI gene ID
HGNC symbol

Dataset

[None Selected]

Export all results to

File

TSV

☐ Unique results only

Go

Email notification to

View

10

rows as

HTML

☐ Unique results only

Gene type	GOSlim GOA Access	GOSlim GOA Description	NCBI gene ID	HGNC symbol
protein_coding	GO:0008150	biological_process	4363	ABCC1
protein_coding	GO:0005575	cellular_component	4363	ABCC1
protein_coding	GO:0022857	transmembrane transporter activity	4363	ABCC1
protein_coding	GO:0003674	molecular_function	4363	ABCC1
protein_coding	GO:0055085	transmembrane transport	4363	ABCC1
protein_coding	GO:0006810	transport	4363	ABCC1
protein_coding	GO:0043167	ion binding	4363	ABCC1
protein_coding	GO:0005623	cell	4363	ABCC1
protein_coding	GO:0005886	plasma membrane	4363	ABCC1
protein_coding	GO:0016887	ATPase activity	4363	ABCC1

Download the results

- Check **unique results only**
- Choose the output format (e.g. XLS)
- Click on **Go** to download results

The screenshot shows the Ensembl BLAST/BLAT interface. The top navigation bar includes links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog. A search bar is on the right. Below the navigation bar, there are tabs for New, Count, and Results. The Results tab is active, showing a sidebar with Dataset (Human genes (GRCh38.p13)), Filters, and Attributes. The main content area has an 'Export all results to' section with a dropdown menu set to 'File', a dropdown menu set to 'XLS', a checkbox for 'Unique results only' which is checked, and a 'Go' button. Below this is a 'View' section with a dropdown menu set to '10', a dropdown menu set to 'HTML', and a checkbox for 'Unique results only' which is unchecked. A table of results is displayed below the View section.

Gene type	GOSlim GOA Accession(s)	GOSlim GOA Description	NCBI gene ID	HGNC symbol
protein_coding	GO:0008150	biological_process	4363	ABCC1
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File Home Inserisci Layout di pagina Formule Dati Revisione Visualizza Sviluppo					General				
Incolla	Taglia	Copia	Copia formato	Appunti	Carattere	Allineamento	Unisci e allinea al centro	Testo a capo	Formattazione condizionale
A1	Gene type								
	A	B	C	D	E				
1	Gene type	GOSlim GOA Accession(s)	GOSlim GOA Description	NCBI gene ID	HGNC symbol				
2	protein_coding	GO:0008150	biological_process	4363	ABCC1				
3	protein_coding	GO:0005575	cellular_component	4363	ABCC1				
4	protein_coding	GO:0022857	transmembrane transporter activity	4363	ABCC1				
5	protein_coding	GO:0003674	molecular_function	4363	ABCC1				
6	protein_coding	GO:0055085	transmembrane transport	4363	ABCC1				
7	protein_coding	GO:0006810	transport	4363	ABCC1				
8	protein_coding	GO:0043167	ion binding	4363	ABCC1				
9	protein_coding	GO:0005623	cell	4363	ABCC1				
10	protein_coding	GO:0005886	plasma membrane	4363	ABCC1				
11	protein_coding	GO:0016887	ATPase activity	4363	ABCC1				
12	protein_coding	GO:0005576	extracellular region	4363	ABCC1				
13	protein_coding	GO:0043226	organelle	4363	ABCC1				
14	protein_coding	GO:0005615	extracellular space	4363	ABCC1				
15	protein_coding	GO:0006950	response to stress	4363	ABCC1				
16	protein_coding	GO:0044281	small molecule metabolic process	4363	ABCC1				
17	protein_coding	GO:0051186	cofactor metabolic process	4363	ABCC1				
18	protein_coding	GO:0061024	membrane organization	4363	ABCC1				
19	protein_coding	GO:0040011	locomotion	4363	ABCC1				
20	protein_coding	GO:0048870	cell motility	4363	ABCC1				
21	protein_coding	GO:0005575	cellular_component	26027	ACOT11				
22	protein_coding	GO:0005623	cell	26027	ACOT11				
23	protein_coding	GO:0005622	intracellular	26027	ACOT11				
24	protein_coding	GO:0005737	cytoplasm	26027	ACOT11				
25	protein_coding	GO:0005829	cytosol	26027	ACOT11				
26	protein_coding	GO:0003674	molecular_function	26027	ACOT11				
27	protein_coding	GO:0043226	organelle	26027	ACOT11				
28	protein_coding	GO:0005739	mitochondrion	26027	ACOT11				
29	protein_coding	GO:0008150	biological_process	26027	ACOT11				
30	protein_coding	GO:0006629	lipid metabolic process	26027	ACOT11				
31	protein_coding	GO:0044281	small molecule metabolic process	26027	ACOT11				